

Supplementary material for “Native CCZ codes from projective geometry: a 48-qubit code and a length lower bound”

Bohan Lu^{1,2} and Kenneth R. Brown^{1,2,3,4}

¹Duke Quantum Center, Duke University, Durham, NC 27701

²Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708

³Department of Physics, Duke University, Durham, NC 27708

⁴Department of Chemistry, Duke University, Durham, NC 27708

This supplement gives exact certificates for the computational results used in the paper and a self-contained analytic proof of the FG48 uniqueness statement.

The paper states the mathematical results and summarizes the finite computations. The supplement gives human-readable certificates for the computational claims that support the obstruction, the parameters and physical signs of $\mathcal{Q}_{48}$, and the length lower bound. The programs and machine-readable outputs are retained in the companion repository. The companion repository’s [reader guide](#) maps each claim below to the corresponding program, output, expected result, and reproduction cost. All searches used for the stated results are exact.

Inventory of computational claims

Note	Claim supported	Form of evidence
A	The $[[47, 3, 3]]$ code satisfies the nine CH conditions but has no native $T/T^\dagger$ pattern	Exact modulo-two contradiction extracted from the native equations
B	$\mathcal{Q}_{48}$ has ranks 9 and 6, distances $d_X = 16$ and $d_Z = 3$, and a $26T + 22T^\dagger$ native pattern	Exact row reduction, complete coset minimization, an explicit Z -logical, and evaluation of all 512 codewords
C	The finite cases in the length lower bound have no admissible label map	Complete support and orbit coverage, followed by exhaustive exact linear-system tests
D	FG48 is the unique projective 8-divisible $[[48, 6]]$ code	Analytic proof included for reference, with no computational dependence

Bohan Lu: bohan.lu@duke.edu

The construction of the FG48 support and the displayed indicator identities also have independent computer checks in the companion repository. The checks confirm displayed formulas whose proofs already appear in the paper. The independent computations are cross-checks rather than additional logical dependencies.

1 Supplementary Note A: the non-native 47-qubit code

Let $g^1, \dots, g^9 \in \mathbb{F}_2^{47}$ be the rows of the generator matrix in Ref. [1], with the first three rows logical and the last six rows stabilizers. A native flip vector $\gamma \in \mathbb{F}_2^{47}$ would satisfy the paper's signed-weight criterion. Reduction of its one-row and two-row equations modulo two gives

$$\gamma \cdot g^i \equiv \frac{|g^i|}{2} \pmod{2}, \quad (1)$$

$$\gamma \cdot (g^i \wedge g^j) \equiv \frac{|g^i \wedge g^j|}{2} \pmod{2}. \quad (2)$$

The overlap in Equation (2) is even because the code satisfies the CH pair conditions.

The following sets of coefficient rows obey one exact dependence over $\mathbb{F}_2$:

$$\begin{aligned} I &= \{3, 4, 8\}, \\ P &= \{(1, 7), (1, 8), (1, 9), (2, 4), (2, 5), (3, 4), (4, 8), (5, 6), (5, 8), (5, 9), (6, 7)\}, \\ \bigoplus_{i \in I} g^i \oplus \bigoplus_{(i,j) \in P} (g^i \wedge g^j) &= 0. \end{aligned}$$

Taking the same linear combination of Equations (1) and (2) makes the left hand side zero. Every row indexed by I has weight 24, so the three single-row terms on the right hand side vanish modulo two. The pair-overlap weights, in the order displayed in P , are

$$14, 14, 12, 12, 10, 8, 12, 12, 12, 12, 12.$$

Exactly three half-weights are odd. The right hand side is therefore one modulo two, which gives the contradiction $0 = 1$. No binary flip vector can satisfy the native equations. Direct row reduction independently gives $\text{rank } G = 9$, $\text{rank } S = 6$, $d_X = 16$, and $d_Z = 3$, and direct evaluation confirms the nine CH conditions.

2 Supplementary Note B: parameters and physical signs of $\mathcal{Q}_{48}$

For the representative stored with the companion programs, row reduction gives $\text{rank } G = 9$ and $\text{rank } S_{\text{FG48}} = 6$. The seven nonzero cosets of $\text{rowspan}(G)/\text{rowspan}(S_{\text{FG48}})$ have the following exact minimum weights.

Logical label	Minimum Hamming weight
001	18
010	16
011	18
100	16
101	18
110	16
111	18

Thus $d_X = 16$.

An explicit weight-three Z -logical selects the columns

$$(u, w) = (0, (1, 0)), \quad (0, (0, 1)), \quad (0, (1, 1)).$$

The three stabilizer labels sum to zero. The corresponding logical labels from the paper's label map are $(0, 1, 1)$, $(1, 1, 0)$, and $(0, 0, 0)$, whose sum is $(1, 0, 1) \neq 0$. The selected columns therefore give a vector z with

$$S_{\text{FG48}} z^\top = 0, \quad K z^\top = (1, 0, 1)^\top.$$

Hence z is a nontrivial Z -logical of weight three. Projectivity excludes weights one and two, so $d_Z = 3$.

Evaluation of the polynomial for the flip vector on all 48 columns gives block flip counts 6, 6, 10. Hence 22 sites carry $T^\dagger$ and 26 sites carry T . Exhaustive evaluation of the signed-weight congruence over all $2^9 = 512$ codewords gives

$$\text{sw}_\Gamma(K^\top x + S_{\text{FG48}}^\top y) \equiv 4x_1x_2x_3 \pmod{8}$$

for every $x \in \mathbb{F}_2^3$ and $y \in \mathbb{F}_2^6$.

3 Supplementary Note C: exact finite checks for the length lower bound

The finite part of the lower-bound proof fixes a syndrome support $\mathcal{P} \subseteq \mathbb{F}_2^8$ and writes each of the three logical labels as a vector in

$$V_{\mathcal{P}} = (\text{RM}(2, s)|_{\mathcal{P}})^\perp.$$

For every admissible pair of the first two label vectors, the remaining CH conditions form a linear system for the third. Each reported outcome is obtained by exact arithmetic over $\mathbb{F}_2$. No random sampling supplies a table entry.

For $s = 5$, the 62 affine-flat and punctured-flat supports fail the flat obstruction. On the remaining support $\mathbb{F}_2^5 \setminus \{0\}$, the search tests all 49,741,825 admissible pairs of the first two label vectors. Every resulting system for the third label vector is inconsistent.

For $s = 6$, the balanced quadratic functions have the three affine normal forms [2, Ch. 15, Thms. 4–5]

$$y_6, \quad y_6 + y_1y_2, \quad y_6 + y_1y_2 + y_3y_4.$$

The first support is the affine hyperplane $y_6 = 1$. Its stabilizer row S_6 is all ones, so adding S_6 to the logical rows makes their entries at $(0, 0, 0, 0, 0, 1)$ zero. Such row additions preserve the CH conditions by the overlap expansions in Appendix D.2. Delete S_6 , then delete the zero full column. The remaining stabilizer columns are exactly $\mathbb{F}_2^5 \setminus \{0\}$, with the CH conditions unchanged. K3 and the stabilizer-containing conditions keep the logical rows independent modulo the stabilizers. The rank-five exclusion therefore applies. For the punctured case, translating the puncture point to the origin gives this rank-five support directly.

To cover every origin choice for the other two forms, write $u = (y_1, \dots, y_5)$, $z = y_6$, and $c(u, z) = z + q(u)$, where $q(u) = u_1u_2$ or $u_1u_2 + u_3u_4$. For an origin choice (a, z_0) , translate that point to zero and apply

$$(u, z) \mapsto (u, z + q(u + a) + q(u) + q(a)).$$

The map is linear and is its own inverse. On the translated support, its output satisfies $z + q(u) = 1 + c(a, z_0)$. Consequently every origin outside the support gives the same unpunctured representative, and every origin inside gives the same punctured representative after zero is removed. Linear coordinate changes preserve the CH equations. The retained verification checks all origin choices and the transported label spaces, not sampled invariant classes.

For each of these representatives, the exact search enumerates every admissible ordered pair of nonzero logical labels and checks the remaining linear equations. K3 excludes a zero logical label, so omitting zero loses no solution.

q	Unpunctured		Punctured	
	$\dim V_{\mathcal{P}}$	Pairs tested	$\dim V_{\mathcal{P}}$	Pairs tested
u_1u_2	13	12,042,241	12	3,006,465
$u_1u_2 + u_3u_4$	11	2,224,129	10	555,009

Every third-label system is inconsistent. Together with the affine-hyperplane reduction, these tests settle the entire rank-six weight-32 case without using the sufficient counting inequality of Appendix D.2.

For $s = 6, 7, 8$, the candidate supports and final outcomes are summarized below. A dash records a case settled before the third-label test by the obstruction named in the last column.

s	$ c $	Support classes or pair orbits	Unpunctured	Punctured	Result
6	16	—	—	—	flat obstruction
6	24	1	1	24	no third label
6	28	1	1	28	no third label
6	32	3	2	2	no third label
6	36	1	1	36	no third label
7	16	—	—	—	flat obstruction
7	24	1	1	24	no third label
7	28	2	2	56	no third label
7	32	4	3	96	no third label
7	36	9,792	7,038	352,512	no third label
8	32	108	13,632	436,224	no third label
8	34	108	3,712	126,208	no third label
8	36	2,399	292,128	10,516,608	no third label
8	38	2,399	160,256	6,089,728	no third label

The unpunctured rank-six weight-36 case tests 43,450,369 admissible pairs. Across the rank-eight rows, 17,638,496 systems reach the label-map tests. Condition K3 rejects 13,274,528 systems before the third-label solve. The remaining 4,363,968 systems are inconsistent.

The rank-eight calculation first fixes the first restriction to a normal form and then reduces the second restriction under the invertible affine maps that preserve the first. The orbit checks give the following complete partitions.

Restriction weights	Functions in the class	Orbits	Sum of orbit sizes
(8, 8)	11,160	20	11,160
(8, 12)	1,749,888	88	1,749,888
(12, 12)	1,749,888	2,167	1,749,888
(8, 14)	22,855,680	124	22,855,680

For every row, the retained orbit sets are disjoint, remain in the stated class, and have total size equal to the full class size. The 2,399 representatives in the two rows that use the complete rank-eight list were all included. The separate sweep of the 124 weight-14 representatives decided 840,384 systems. The output contains no missing or deferred case and no surviving system. Together with the reductions proved in the paper, these checks cover every candidate of length at most 38.

4 Supplementary Note D: analytic uniqueness of FG48

Proposition 4.1 (The $[48, 6]$ stabilizer code is forced [3, 4]). *The only projective 8-divisible $[48, 6]$ code is the two-weight complement of a solid on $\text{PG}(5, 2) \setminus \text{PG}(3, 2)$. It is the X -stabilizer subcode S_{FG48} of Section 4.*

Proof. Identify the code with its set X of 48 distinct columns (projective: distinct and nonzero) in $\text{PG}(5, 2)$. The ambient space has 63 points. Each nonzero codeword c_a corresponds to a hyperplane H_a of 31 points, and its weight counts the points of X the hyperplane misses, $|c_a| = 48 - |X \cap H_a|$. Eight-divisibility therefore says that every hyperplane meets X in a multiple of eight points. A hyperplane misses only 32 of the 63 ambient points, so it contains at least 16 of the 48 points of X , and at most all 31 of its own. The only multiples of eight in that range are 16 and 24, giving weights 32 and 24. Writing h_{16}, h_{24} for the numbers of hyperplanes meeting X in 16 and 24 points, and double-counting hyperplanes and point-hyperplane incidences (each of the 48 points lies in 31 hyperplanes),

$$h_{16} + h_{24} = 63, \quad 16h_{16} + 24h_{24} = 48 \cdot 31,$$

a determined system with the unique solution $h_{16} = 3, h_{24} = 60$. The weight enumerator is thus forced to be $\{32:3, 24:60\}$. It remains to show this enumerator has a unique projective realization.

Form the Cayley graph on the group $\mathbb{F}_2^6$ with connection set X (symmetric since $-x = x$). Its eigenvalues are the character sums

$$\lambda_a = \sum_{x \in X} (-1)^{a \cdot x} = 48 - 2|c_a|,$$

namely 48 once (at $a = 0$), -16 for the three weight-32 words, and 0 for the sixty weight-24 words. Passing to the complement connection set $Y = (\mathbb{F}_2^6 \setminus \{0\}) \setminus X$ of 15 points shifts each nontrivial eigenvalue to $-1 - \lambda_a$, so the complement graph is 15-regular with spectrum in $\{15, -1\}$, and its adjacency matrix satisfies

$$(A - 15I)(A + I) = 0, \quad \text{i.e.} \quad A^2 = 14A + 15I.$$

Adjacent vertices therefore share all fourteen remaining neighbours: every neighbourhood is a clique, and the graph is a disjoint union of complete graphs on 16 vertices. Components of a Cayley graph are the cosets of the subgroup generated by the connection set, so the component of 0 is a 16-element subgroup $P \subseteq \mathbb{F}_2^6$ with $Y = P \setminus \{0\}$. Hence $X = \mathbb{F}_2^6 \setminus P$ is the complement of a four-dimensional subspace, the deleted solid. $\square$

Delsarte's correspondence associates a projective two-weight code with a strongly regular Cayley graph on its connection set. The graph of FG48 is complete multipartite, with parameters $\text{SRG}(64, 48, 32, 48)$ and parts given by the cosets of P . The three weight-32 words are the characters trivial on P . Subspace complements sit at the imprimitive end of the correspondence, which explains why uniqueness holds here while generic two-weight parameters admit many inequivalent realizations [4, 5].

References

- [1] Hugo Jacinto, Xavier Valcarce, Victor Barizien, Élie Gouzien, and Nicolas Sangouard. Exploring the landscape of compact magic-state distillation factories, 2026. arXiv:2606.07734.
- [2] Florence J. MacWilliams and Neil J. A. Sloane. *The Theory of Error-Correcting Codes*. North-Holland, 1977.
- [3] Sascha Kurz. Divisible codes. 2021. arXiv:2112.11763.
- [4] A. R. Calderbank and W. M. Kantor. The geometry of two-weight codes. *Bulletin of the London Mathematical Society*, 18(2):97–122, 1986.
- [5] A. E. Brouwer. Two-weight codes. In W. C. Huffman, J.-L. Kim, and P. Solé, editors, *Concise Encyclopedia of Coding Theory*. CRC Press, 2021.